Tutorial 25 — LLM-Driven Evolutionary Search

Chapter 10 with two AlphaEvolve circle-packing problems

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1 Agentic evolutionary frameworks

Today we will build the Chapter 10 evolutionary loop around a concrete mathematical task. We evolve programs that search for circle packings. The circles are the programs' outputs; they provide evidence about program quality. We will use both unit-square examples from AlphaEvolve, with 26 and 32 circles. Their mathematical problems are reproduced; our short, local Luna experiment is a classroom implementation of the discovery loop, not the unpublished AlphaEvolve engine or a claim to reproduce its reported records.

2 10.1 The evolutionary loop

2.1 An external evolutionary loop

What must survive between calls for the next attempt to build on the previous one? Keep executable candidates, their measured results, and the changes already tried. Select survivors before proposing their descendants.

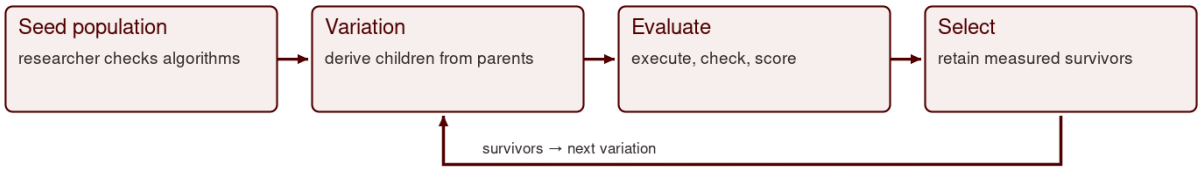


Figure 1: Initialize once; evaluate, select and mutate repeatedly.

Evolution gives the next proposal ancestry and external memory. Repeated independent generation does not inherit the survivors.

Chapter 10.1 distinguishes an evolutionary loop from the generate–evaluate–select loop of Chapter 9: mutation consumes the survivors. Initialization happens once. Later rounds preserve an artifact and vary its structure instead of asking for an unrelated answer every time. The outside controller stores programs and measurements; this memory persists even though the language model's weights stay fixed. An external evaluator supplies evidence beyond a model's verbal self-assessment.

A survivor is a program. Executing it produces a circle arrangement. Keeping a successful arrangement can also provide a numerical starting point for later programs, but that is separate from inheriting program source. Our default demonstration exposes both: program ancestry and a frozen bank of the previous best packings.

2.2 Five choices define the search

Which choices turn "try to improve this" into a reproducible procedure?
$(\mathcal{A}, f, \mu, \sigma, P_0)$: program space, fitness, mutation, selection, and initial population.

$$P_{t+1} = \sigma(P_t \cup \{\mu(a) : a \in \sigma(P_t)\}) .$$

Choose parents → mutate → pool parents and children → select survivors.
The inner selection chooses parents; the outer selection forms the next generation. Every component is a design choice.

This is Chapter 10, Definition 10.2 and Equation 10.1. The course minimizes fitness f. For our maximization problem, use negative median verified sum of radii. The artifact space consists of Python packing programs with a fixed function interface and bounded execution. Mutation is a Luna request containing actual selected parent source. Selection retains at most three complete programs. Checkboxes choose the initial population; they do not restrict the source changes Luna may propose.

The displayed recurrence is the chapter's conceptual definition. Our controller sends up to three selected parents in one request and asks for one child, rather than making one request for every parent. It then evaluates that child and selects from parents plus child. This is a declared classroom choice, not a hidden implementation of every feature of AlphaEvolve. The chapter's fixed fitness f corresponds directly to our fixed-start control. Default iterative refinement uses conditional fitness f(a | B_t), where the incoming packing bank is explicit evaluation state. Re-evaluating parents makes each round comparable; it does not make medians from different banks stationary program quality.

3 10.2 Mutation operators

3.1 Classical variation

How can we vary a program without understanding why it works?
Point mutation: change a move scale, for example $0.04 \to 0.02$. **Subtree replacement:** replace an acceptance expression. **Crossover:** inherit code from two parents.
These edits create hypotheses. The evaluator determines whether each resulting program still runs and improves its measured fitness.
Classical variation changes structure; selection supplies direction. Small source edits can have large behavioral effects.

Chapter 10.2.1 introduces point mutation, subtree mutation and crossover using expression trees. Translate that vocabulary to the code students will see. A literal change is a point edit. Replacing an acceptance rule changes a component of the computation. Combining one parent's constructor with another parent's search schedule is inheritance from multiple programs. These are conceptual examples, not claims that our controller implements a classical genetic-programming engine.

A move scale controls the distance of a proposed center displacement. Changing it does not by itself say whether a program is better. A smaller scale can help refine a packing but may impede escape from a basin. The complete program must be run and checked under the same trial conditions. This sets up the difference between blind variation and a mutator that reads numerical feedback.

3.2 A language model writes the mutation

What can a mutator change when it reads the parent code and its results?

1. Retain the parent's center search and radius allocator.
2. Propose a different move scale, center subset, or acceptance schedule.
3. Luna returns edited **Python source** and a hypothesis.
4. An isolated interpreter executes it; an independent checker evaluates the returned circles.

Luna implements the proposed code change; Python executes it. A reasoned edit becomes evidence only after evaluation.

Chapter 10.2.2 replaces blind variation with knowledge-guided variation. For example, a parent may perturb one center at a time and stop improving. The mutator can hypothesize that coordinated moves or a restart schedule will escape that basin. It writes the new function; the local Python interpreter does not translate the prose hypothesis into an algorithm.

Our response contains round_id, parent_ids, hypothesis and source. The source defines solve(initial, seed, steps). NumPy, math, a fixed-center radius allocator and a progress reporter are supplied. A source diff exposes what was retained and changed. The geometry checker and survivor rule are outside the editable source. Bubblewrap isolates generated code from credentials, private repository files and the host network. The model has no execution tools during its proposal call.

AlphaEvolve evolves code and evaluates it externally (AlphaEvolve team 2025). We use the published circle-packing tasks with our own small controller and TAMU Luna. The public results repository does not release the original discovery engine. Do not describe our initial numerical methods as the algorithms AlphaEvolve discovered.

3.3 Context joins the two course spines

Which evidence should the next mutation see?

Parent source **and scores**; successful and failed attempts; task constraints and useful domain knowledge; population similarity and remaining budget.

Evaluation becomes fitness. Context engineering drives mutation. The next request carries the previous round's actual results.

External memory makes the next proposal informed. A plausible explanation does not replace a measured trial.

These are the four context ingredients in Chapter 10.2.3 and the convergence of the two spines in Figure 10.2. Our domain knowledge includes the nonoverlap constraints and the fact that radius allocation is a linear program once centers are fixed. Source, scores, failures and tried IDs expose what the search already knows. The context labels the current packing bank, so scores from different starting conditions are not silently mixed.

The server's request panel shows the actual provider payload; the response panel shows the public stream. Authorization is added separately inside the model worker. Hidden reasoning is neither requested as teaching evidence nor displayed. Numerical seeds control starting packings and numerical moves. They do not determine Luna's wording. The optional structure-only control withholds scores and history, but a rigorous context comparison must additionally freeze identical parent programs and histories across the two conditions.

4 10.3 Selection and population management

4.1 Elitism protects measured progress

If a mutation is worse, must the current champion disappear?
Pool parents with the new child. Retain the best complete programs under the declared fitness rule.
For a maximized quality, elitism makes the observed best stay level or improve **under comparable evaluation conditions**.
A failed mutation need not erase a useful parent. Preserve the best valid packing separately from program fitness.

Chapter 10.3.1 describes carrying elite candidates forward. The lower-is-better course fitness is monotonically non-increasing when the same deterministic measured fitness is used. Equivalently, maximized observed quality is non-decreasing. This is a property of retaining a measured incumbent, not a theorem of global convergence or future quality on new instances.

Our default uses inherited packing banks. At each round, freeze the same incoming packings for the parents and the child, re-evaluate the parents, and select within that bank. An old program median from a different bank is not directly comparable. The cumulative best valid construction is a separate monotone curve because its verified geometry is retained. A valid progress packing can survive in that archive even when a program later times out and receives no eligible fitness.

4.2 Selection pressure and population collapse

Why can keeping only the highest scores make search stagnate?
Top-k: rank everyone and retain the best k. **Tournament:** sample a group and retain its best; smaller tournaments can preserve more alternatives.
Collapse: several survivors become small rewrites of the same search. Current fitness does not reveal every parent's future descendants.
Strong selection exploits evidence but can discard useful routes. Our top-three rule does not guarantee diversity.

Chapter 10.3.2 compares top-k and tournament selection; 10.3.3 identifies population collapse. Our implemented rule is deterministic: all three trials must complete and pass geometry checks; then rank by median sum of radii, minimum sum, and stable source identity. A child below the champion may still survive if it belongs among the three best programs. Do not teach elitism as a rule that rejects every nonchampion.

Programs with different IDs may have nearly identical behavior. Normalizing syntax prevents comments alone from creating another genome, but it is not a semantic diversity measure. Several schedules differing only in a constant can still explore the same region. The guide names our three initial mechanisms so students can inspect their differences before deciding whether the population is varied enough. Tournament selection is explained here but is not implemented as a hidden classroom feature.

5 10.4 Fitness design

5.1 Feasibility comes before fitness

Could a flawed scorer reward an invalid packing?
Flawed scorer: sum radii without checking overlap. A valid pair scores **0.50**; an overlapping pair scores **0.60** and incorrectly wins.
Corrected evaluator: check every wall and pair, then score. Reject invalid or incomplete trials.

$$f(a \mid B_t) = -\operatorname{median}_{j=1,2,3} S(a; z_{t,j}).$$

Compare programs within one frozen packing bank B_t.
Close the scoring loophole; then test frozen programs on unseen starts. Feasibility and transfer are separate checks.

Chapter 10.4 treats the evaluator as the fitness function and warns that persistent optimization finds weaknesses in it. Here we explicitly compare a flawed scorer with the corrected evaluator, rather than merely presenting an invalid output. The flawed scorer returns the sum of radii and omits pair checks. It ranks two circles of radius 0.30 at centers (0.30, 0.50) and (0.70, 0.50) above two radius-0.25 circles at (0.25, 0.50) and (0.75, 0.50). The claimed gain is 0.10, but the first pair overlaps by 0.20. An optimizer rewarded by that flawed rule has an incentive to enlarge overlapping circles. The correct geometry checker rejects the apparent winner. These are instructor-made, numerically checked toy examples, not Luna outputs. The algorithm guide contains a revealable scorer illustration.

A production witness must contain exactly the required number of finite circle triples. Radii must be nonnegative, circles must fit inside the square, and every pair must avoid overlap. The server independently recomputes both geometry and summed radii; it ignores a program's claimed score. Numerical tolerance is 1e-9 in unit-square units. A common radius-allocation construction margin of 1e-8 is applied before a final witness is returned and scored. No invalid output is silently repaired during evaluation.

Fitness is the negative median quality of three complete trials on the same bank. A failed trial is not quietly dropped from the average; it makes the program ineligible for selection. A valid intermediate packing is still a separate useful construction. This distinction is essential for anytime numerical search.

Feasibility checking does not establish algorithm transfer. Freeze program source before evaluating fresh numerical starts, exclude those validation results from mutation prompts and survivor decisions, and report them separately. The original runs used seeds 3, 4 and 5. A separate numerical-only check uses previously unused seeds 101, 102 and 103, common fixed starts, the same requested work and the same hard trial limit. The selection rule was fixed before seeing validation scores: compare the best initial program with the highest-ranked surviving new Luna program. The guide shows the actual results and source identities. This is a small post-search check, not a broad generalization study or a second live evolution.

Returning a fixed valid construction can be a legitimate solution to a fixed packing problem; a hard-coded layout alone is not proof of reward gaming. The loophole here is accepting infeasible geometry. Overfitting claims concern performance outside the observed starts or problem instances and require a separate evaluation contract. Multiobjective quality and cost are developed in the exploration frame and guide. The changing center positions define a nonconvex landscape; the

fixed-center radius linear program does not solve that whole problem.

6 10.5 Diversity and exploration

6.1 Preserve different search behaviors

How can useful alternatives survive beside the current champion?
Islands: evolve separate populations, with occasional migration. **MAP-Elites:** retain the best program in each behavior bin, such as one-center versus coordinated moves.
Novelty search: favor behaviors unlike those already seen. Different source text alone does not establish different search behavior.
Preserve several ways to search. These are possible extensions; our top-three controller does not implement a diversity archive.

Chapter 10.5.1 motivates preserving several search histories. An island owns a population and mutation history. Occasional migration shares programs; overly frequent migration can make the islands alike, while too little sharing can leave each stalled. Three evaluation seeds are not three islands, and the two circle counts are different problems rather than populations exchanging discoveries.

The Multi-dimensional Archive of Phenotypic Elites (MAP-Elites) organizes candidates by chosen behavior descriptors and keeps a high-quality representative in each bin (Mouret and Clune 2015). For a packing-program archive, illustrative descriptors could measure the typical number of centers moved per proposal and the frequency of restarts. A one-center local search and a coordinated-move search could occupy different bins even if one currently has a lower verified sum. Their quality would still need comparable evaluation conditions. Those descriptors would require instrumentation; the present worker does not record them as a ready-made archive.

Novelty search instead scores behavioral difference from previously observed behaviors (Lehman and Stanley 2011). A possible descriptor distance could use normalized numerical search statistics. It is not a measure of how unusual a program's variable names sound, and novelty alone does not certify packing quality. These are alternative selection designs whose performance depends on the task and descriptors.

Our controller retains the top three complete programs and rejects exact syntax duplicates. It has no islands, behavioral bins or novelty score. The guide works through the conceptual archive example so students can recognize both its purpose and its absence from the bounded demo.

6.2 Exploration, exploitation and cost

When should we refine, explore—or stop spending?
Exploit: refine. **Explore:** change methods. **Stop:** respect the cap.
Hypothetical policies, same test conditions:

Policy	Median sum	Numerical seconds
A	2.40	1
B	2.41	5

Illustrative utility $S - 0.01t$ favors A: $2.39 > 2.36$**.** Our demo reports costs separately.

Chapter 10.5.2 frames the explore–exploit tension. Exploitation refines useful inherited structure;

exploration tests a different search mechanism. Internal numerical exploration, such as accepting temporary losses, is different from an outer mutation changing the program. Neither a large source edit nor a fluent hypothesis guarantees an improvement.

The table is invented to explain a decision, not measured Luna performance. Both policies must first be valid and evaluated comparably. B offers 0.01 more quality but takes four more numerical seconds. Neither dominates: A is faster and B has higher quality. A quality–time Pareto view therefore keeps both as options. If we explicitly choose a value of 0.01 quality units per second, maximizing utility U=S−0.01t gives A=2.39 and B=2.36. The price is a teaching assumption, not a measured API tariff or universal preference. A smaller price could reverse the decision.

Chapter 10.8 expresses the corresponding minimization form as cost-adjusted fitness. Cost may mean marginal API dollars or a declared resource measure, with a coefficient that converts units. Generation cost belongs to the path that produced a program; it is not an intrinsic property of source code. Avoid double-counting shared preparation and distinguish per-program evaluation time from all model calls and numerical effort spent finding it. Source length is not automatically a good proxy for either runtime or algorithm quality.

The implemented demo keeps quality primary and reports numerical effort and model usage separately. It does not apply the illustrative price, a complexity penalty or a Pareto archive. Three model calls, 35 seconds per call, eight seconds per trial and a 240-second run deadline bound work. An optional predeclared plateau stopping rule could save further requests; it is not the current automatic policy. The current run can continue its planned rounds after a plateau so students can inspect the mechanism.

No number of calls is universally optimal. Three calls per demonstration are a teaching budget: the first child, a proposal informed by its evaluation, and another chance to inspect adaptation or plateau. This budget is far smaller than the research-scale searches in the AlphaEvolve report. A flat or failed run remains useful evidence. Do not add artificial motion or repeated requests to force a desired trajectory.

7 Two published circle-packing problems

7.1 Demo 1: the 26-circle problem

What exactly makes a packing valid, and what does "better" mean?
Place $N = 26$ variable-radius circles in the unit square and maximize $\sum_i r_i$.

$$r_i \geq 0, \quad r_i \leq x_i, y_i \leq 1 - r_i, \quad \|c_i - c_j\|_2 \geq r_i + r_j.$$

These are AlphaEvolve's published constraints and objective. We search live with Luna; the paper's reported construction is a separate research result.

Appendix B.12 of the AlphaEvolve report gives this unit-square, sum-of-radii problem for 26 and 32 circles (AlphaEvolve team 2025). This is variable-radius packing; it is not equal-radius packing and not maximization of covered area. Tangency and zero radii are allowed under our stated contract; zero-radius count remains visible. The paper reports historical construction scores, not certificates of optimality, and they are not hidden inputs to our search.

Start the first demo with 26 selected in the instance control. Show the three initial Python programs in the guide: perturbed grid construction, greedy local center search, and simulated annealing of

centers. The fixed-center radius allocator is common. The constructor intentionally spends less effort; compare actual numerical costs as well as quality. All programs in a round receive copies of the same packing bank for seeds 3, 4 and 5.

7.2 Demo 1: follow three mutations

Which part of the parent survives, and does the changed program earn its place?
Evaluate the initial population → inspect parent source and context → send Luna round 1 → read the actual code diff and verified packing.
Repeat for rounds 2 and 3. Inspect inherited packing-bank IDs, complete trial scores, failures and survivor decisions.
Follow the real chain: source → execution → checked geometry → fitness → selection → next context.

Open the integrated demo from this frame. Select LIVE LLM, 26 circles, three calls, mutation, evidence-rich context and inherited packings. Check the selected initial programs and evaluate them. This starts numerical work only; every model call waits at an explicit gate.

At the first gate, point to the incoming packing bank: it contains the best valid construction found for each seed. Parents are re-evaluated from that bank before their scores enter the context. When sending a round, identify the actual provider request, public response and source string. Show who implemented the proposal: Luna wrote the Python; the isolated interpreter executes that exact admitted source. A live intermediate packing is independently verified, but it is not yet complete program fitness.

After evaluation, compare all three trials and the actual top-three decision. Before sending the next call, show that the latest results have entered the context and the packing bank has been updated. Keep failures or plateaus visible. Use the remaining teaching time for questions and inspection if computation finishes early; no artificial delay is inserted. A genuine recording may be used when needed, with a persistent RECORDED REPLAY label.

7.3 Read the first run's evidence

What improved: one packing, the program's measured fitness, or both?
A witness proves one feasible construction. Three complete trials rank a program on one declared bank. A source diff explains what changed; it does not prove an improvement.
Keep construction progress and program comparison separate. Report the actual outcome, calls and numerical effort.

Ask students to point to the record supporting each claim. The circle view includes program identity, seed, validity and summed radii. The table includes complete or failed trials, medians and current-bank selection. The cumulative construction curve retains valid witnesses; it must not be mislabeled as a fixed-protocol program-fitness curve when banks change.

Do not substitute expected improvements for measured outcomes. If an eligible child is worse than the champion, it can still enter another slot. If a trial fails, the program has no eligible median. Matching a historical literature score would still not prove optimality. A three-call classroom run cannot establish broad model or algorithm superiority. The guide's separate numerical validation illustrates how to test frozen programs beyond the evolution starts, rather than treating geometric feasibility as proof of transfer.

7.4 Demo 2: the 32-circle problem

Does the same program-evolution mechanism still work when the number of circles changes?
Use $N = 32$, the second unit-square case in AlphaEvolve Appendix B.12. Keep the objective, checker, trial seeds, source interface and numerical limits declared.
Change the problem instance, then inspect adaptation. A higher raw sum for 32 circles is not evidence of a better algorithm than the 26-circle run.

Reset and select 32 circles. The problem fingerprint changes, as do the number of wall and pair constraints. The method signatures accept n=len(initial), so a program must not assume 26 rows. This is a second published mathematical problem, not an arbitrary graph substitution. Its score scale is different; do not compare raw sums between the two circle counts as though they measured the same instance.

Use the same three disclosed initial methods and seeds so students can recognize the mechanism, but do not call this a controlled test of generalization. The programs are evolved with access to each task. The point is to observe the same Chapter 10 components operating on a changed instance. A separate withheld-size evaluation would be required for a transfer claim. A separate fixed-start check of already-frozen programs is now available in the algorithm guide. It used new seeds 101–103, zero additional model calls and no feedback into evolution. The initial baseline and a surviving new program were selected before inspecting those results. This small check concerns fresh starts at the same N; it is not an unseen-size transfer study.

7.5 Demo 2: inspect adaptation and limits

Does Luna change its search after seeing the 32-circle trial evidence?
Evaluate 32-circle parents → send three gated requests → inspect actual source edits, verified packings and the feedback entering each next request.
Keep invalid proposals, incomplete executions and repeated ideas visible. Stop cancels owned work; replay remains explicitly recorded.
The mechanism is testable even when improvement is absent. Evidence decides which proposed programs survive.

Follow the same procedure as demo one. Make the changed n visible in the request context and in the circle count of returned witnesses. A response that hard-codes the wrong number is invalid; the interpreter does not silently fix it. A source syntax error, duplicate, invalid packing or hard timeout remains part of the outcome and does not trigger an unannounced free retry.

The class should see the proposal arrive, the function execute and the resulting geometry checked. Compare the child with parents evaluated from the same incoming packing bank. Inspect the actual next request rather than asserting that the model learned. Its weights remain unchanged; its context contains the search memory. If a recorded run is used, show its source identity and recorded timing, while making clear that no new request is being made.

7.6 Name the framework we observed

Can you identify every Chapter 10 component in the live record?
$\mathcal{A}$: Python packing programs. f: verified current-bank trial fitness. μ: Luna's code mutation. σ: complete-program selection. P_0: checked initial programs.
Context proposes; external evaluation measures; selection preserves. For your research problem, define these five choices before spending search budget.

Return to the tuple after both demos so its components are now concrete. Students should distinguish the evolving program, the numerical packing returned by execution, and fitness used to select programs. They should also be able to explain ancestry, inherited numerical starts, context feedback and evaluator independence.

Chapter 10 supplies the conceptual structure. AlphaEvolve supplies the two published problem formulations and a real precedent for evolving search programs. Our bounded implementation supplies inspectable classroom evidence. The mathematical task matches the paper; the original engine, model ensemble, compute scale and reported optimized constructions are not reproduced. Those limits matter when explaining what the demonstration establishes.

A Appendix: architecture and data flow

A.1 One server coordinates the experiment

Who owns the population, execution and measurements?

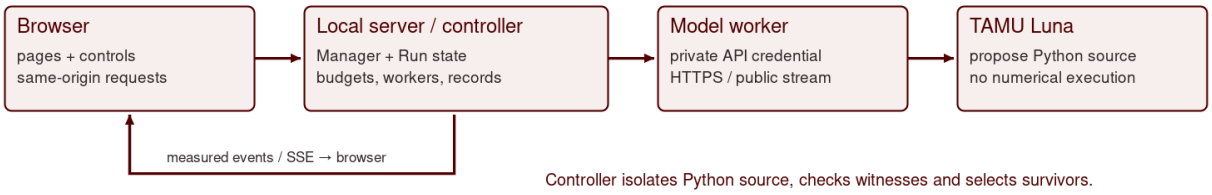


Figure 2: The server coordinates isolated execution and a separate provider worker.

The controller owns the loop. Only the model worker contacts TAMU; a separate isolated worker executes generated Python.

The same loopback server serves the presentation, demo, algorithm guide, exchange, notes and PDFs. Browser controls create a run, start numerical evaluations and explicitly release each model gate. Server-sent events expose actual progress and support reconnect without repeating work. The model worker alone receives the configured provider credential. The generated program runs inside Bubblewrap with read-only runtime code, no host network and no repository, run-log or private-home mounts. Geometry is checked again in the server outside that code.

A.2 Request, context and source response

What crosses the model API boundary?
Send the task, parent source, current-bank trial results, history and limits. Authorization is a separate HTTPS header.
Return JSON: `round_id`, `parent_ids`, `hypothesis`, `source`. The source defines `solve(initial, seed, steps)`.
A source string is executable code after admission. The hypothesis is an explanation, never the authoritative score.

The TAMU transport uses protected.gpt-5.6-luna, public streaming and no tools. It sends system instructions plus a JSON response schema and serialized context. The actual body is displayed on the demo page; the key is never part of it. Source identity is normalized syntax plus executor version. An inherited proposal must name actual surviving parent IDs. An independent proposal must use an empty parent list. The controller rejects duplicates and malformed responses without silent retries.

A.3 Execution, checking and feedback

When can a returned program become a survivor?
Validate source → isolated trials → verify every circle and pair → aggregate three complete scores → select within the current packing bank.
Keep exact request/response records, valid witnesses and failures. The next context carries the measured history.
Execution turns a proposal into evidence. Live LLM, numerical-only and recorded replay remain distinct throughout.

Each numerical trial has an eight-second hard wall cap, including its isolated startup. Each model request has a 35-second cap, and the run has a 240-second deadline with bounded owned-process cleanup. Stop and Reset cancel the owned worker tree, not unrelated processes. Missing isolation or dependencies blocks execution instead of falling back to unsafe or fabricated work.

The numerical-only mode evaluates disclosed initial source programs without contacting Luna. Live mode performs real requests only at explicit gates. Replay is registered only from genuine provider records with matching task, source identities, geometry and transcript provenance. It reproduces recorded events, and reveals responses only as its timeline reaches them. Replay is not a new measured result under a changed executor.

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